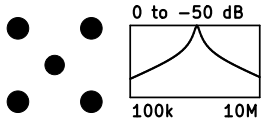
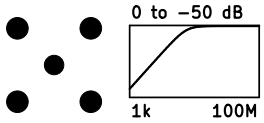
AMOD-P passive building blocks for the converters

Jason Gallicchio and Claude, rev A, 2026-10-25 github.com/gallicchio/AMOD-and-RFMOD

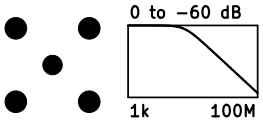
p4 Series RLC
Q about 6
1: +47 pF
2: +100 pF
1.07-0.83 MHz



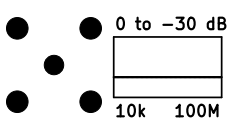
p3 RC high-pass
OUT: 1 k source
ADC on high-Z



p2 RC low-pass
OUT: 1 k source
ADC on high-Z
tau 1 us

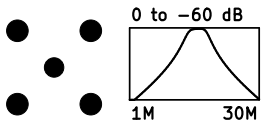


p1 Pi pad
-20 dB, 50 ohm
-14 dB if high-Z
IN max +-3 V

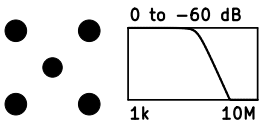


p8 Quartz crystal
fs 4.000 MHz
Q about 50,000
drive < +-1 V
step a few Hz

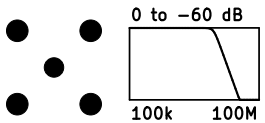
p7 Band-pass
50 ohm both
ends



p6 LC low-pass
50 ohm both
ends



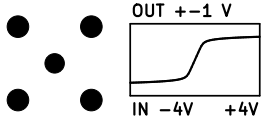
p5 LC low-pass
50 ohm both
ends
-30 dB 18 MHz



p11 Delay line
50 ohm both ends
20 ns a section
taps x1 high-Z
taps /10 50 ohm
cut-off 16 MHz

p9 Ceramic res.
455 kHz series
Q a few 100
one peak, no
flat passband

p10 Clipper
IN max +-4 V
OUT: high-Z



p13 AM loopstick
330 uH coil
0.4-1.3 MHz
VAR C: outside
tuning cap
then a4's LNA

github.com/gallicchio/
AMOD-and-RFMOD
handbook and generator

p12 LEDs
IN 0-5 V
1: 630 nm
2: 596 nm
3: 525 nm
4: 469 nm

AMOD-P: passive building blocks for the FPGA tutorial's converters. Each tile is one circuit with its own SMA jacks: cable DAC OUT to a tile's IN and its OUT to ADC IN. Every jack takes +-4 V. No power. The handbook has each tile's expected response. p13: a 330 uH ferrite-rod coil along this edge, under cable ties; a tuning capacitor on VAR C.

AMOD-P rev A, bottom side seen from below (mirrored), scale 1.90:1